School Management System: PRD

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Overview

School Management System (SMS) ek web aur mobile platform hai jo admissions, attendance, fees, exams aur parent communication ko ek jagah manage karta hai, taake school ka kaam registers aur Excel sheets se nikal kar digital ho jaye.

Problem: Chhote aur darmiyane schools mein records kaghaz ya alag alag files mein hote hain. Is se fee recovery mein deri, attendance aur result mein ghaltiyan, aur parents se rabta kamzor rehta hai. Principal ko school ki asal tasveer (kitne bache present, kitni fee baqi) real time mein nahi milti.

Vision: Ek aasaan, Urdu/English dono mein chalne wala system jo har school ko, bina IT staff ke, apne students, staff aur fees ka poora hisaab saaf aur mehfooz rakhne de.

Assumptions: Ye PRD Pakistan ke private schools (100 se 1,500 students) ko target karta hai. Currency PKR hai, aur fee payment ke liye JazzCash/Easypaisa/bank challan consider kiye gaye hain.

Goals aur Non-goals

Goals

- Fee collection ka waqt aur baqaya (outstanding) 30% kam karna pehle 2 terms mein.
- Daily attendance 5 minute se kam mein mark karna, aur absent hone par parent ko 1 minute ke andar SMS/WhatsApp bhejna.
- Result card aur report generation ka kaam din se ghanton mein lana.
- Principal ke liye ek dashboard: attendance, fee collection aur exam performance ek nazar mein.
- Parents ke liye mobile par fee, attendance, result aur notices dekhna.

Non-goals (is version mein nahi)

- Online classes / video lectures (LMS).
- Library, hostel aur transport ke mukammal modules (sirf transport fee shamil hogi).
- Multi-campus chain ka consolidated ERP.

- Payroll ki mukammal tax filing (sirf salary record aur payslip).

User personas

System chhe qisam ke users ke liye banega; har role ko sirf apna zaruri data nazar aayega.

Role	Kaun hai	Sab se bari zaroorat	Main device
Admin / Owner	School owner ya IT in-charge	Users, classes, fee structure aur settings control karna	Laptop
Principal	Head of school	Poore school ki performance dashboard par dekhna	Laptop, phone
Teacher	Class ya subject teacher	Jaldi attendance, marks entry, homework dena	Phone
Accountant	Fee aur accounts clerk	Fee voucher banana, payment record, baqaya follow-up	Laptop
Parent	Student ka walid ya walida	Bache ki attendance, fee, result aur notices dekhna	Phone
Student	Class 6 se upar ke bache	Timetable, homework aur result dekhna	Phone

Scope aur modules

MVP mein 8 modules hain jo school ka roz marra ka kaam cover karte hain; baqi features Phase 2 aur 3 mein aayenge.

Module	Kya karta hai	Phase
Users aur Roles	Login, role-based access, school settings	MVP
Admissions aur Students	Admission form, student profile, class/section assignment	MVP
Attendance	Daily student aur staff attendance, absent alerts	MVP

Module	Kya karta hai	Phase
Fee Management	Fee structure, voucher, payment record, baqaya, receipts	MVP
Exams aur Results	Exam schedule, marks entry, grades, result card (PDF)	MVP
Communication	Notices, SMS/WhatsApp alerts, parent app notifications	MVP
Dashboard aur Reports	Principal dashboard, attendance/fee/result reports	MVP
Timetable aur Homework	Class timetable, teacher assignment, homework diary	MVP
Parent aur Student App	Native Android/iOS app (MVP mein mobile-friendly parent web portal hoga)	Phase 2
Online Payments	JazzCash, Easypaisa, card payment integration	Phase 2
Staff aur Payroll	Teacher record, leave, salary sheet, payslip	Phase 2
Library aur Transport	Book issue/return, van route aur fee	Phase 3

Functional requirements

Har requirement MVP ke liye hai jab tak Phase na likha ho. Priority: P0 = launch ke liye lazmi, P1 = launch ke baad jaldi.

ID	Module	Requirement	Acceptance criteria	Priority
FR-1	Users	Admin roles aur permissions set kar sake	Har role sirf apni permitted screens dekhe; unauthorised URL par access denied aaye	P0
FR-2	Users	Login phone number/email aur password se, password reset ke saath	Reset link ya OTP 10 minute mein expire ho	P0

ID	Module	Requirement	Acceptance criteria	Priority
FR-3	Admissions	Online/offline admission form se student record banana	Form submit par unique student ID bane; required fields khali hon to save na ho	P0
FR-4	Admissions	Bulk import: Excel se purane students upload	500 students ki file 2 minute mein import ho; ghalat rows ki list download ho	P1
FR-5	Attendance	Teacher class ki attendance ek screen par mark kare	40 bachon ki attendance 2 minute se kam mein save ho; default sab present	P0
FR-6	Attendance	Absent bache ke parent ko automatic alert	Attendance save hone ke 1 minute mein SMS/WhatsApp jaye; delivery status record ho	P0
FR-7	Fee	Class-wise fee structure (tuition, admission, transport, fine) banana	Ek structure se poori class ke vouchers ek click mein ban jayen	P0
FR-8	Fee	Payment record karna: cash, bank, online	Har payment par receipt number aur PDF receipt bane; double entry par warning aaye	P0
FR-9	Fee	Baqaya fee ki list aur reminders	Due date guzarne par parent ko reminder jaye; defaulters ki list export ho	P0
FR-10	Fee	Siblings discount aur scholarship	Discount rule lagne par voucher ka total khud update ho	P1
FR-11	Exams	Exam schedule aur subject-wise marks entry	Teacher sirf apne subject ke marks bhar sake; total aur percentage khud banen	P0

ID	Module	Requirement	Acceptance criteria	Priority
FR-12	Exams	Result card PDF (Urdu/English)	Poori class ke result cards ek saath PDF mein download hon	P0
FR-13	Communication	Notices aur announcements class ya poore school ko	Notice parent portal par aur chune hue channel par jaye	P0
FR-14	Timetable	Class aur teacher timetable, clash detection	Ek teacher ko ek waqt do classes dene par warning aaye	P1
FR-15	Homework	Teacher homework post kare, parent dekhe	Post hone par parent portal par usi din nazar aaye	P1
FR-16	Dashboard	Principal ke liye live summary	Aaj ki attendance %, is mahine ki fee collection aur baqaya 3 second mein load hon	P0
FR-17	Reports	Attendance, fee, result reports export (PDF/Excel)	Date range aur class filter ke saath export chale	P0
FR-18	Parent portal	Parent apne bache ki attendance, fee, result dekhe	Ek parent ke multiple bache ho to switch kar sake; sirf apne bache ka data nazar aaye	P0

Non-functional requirements

Area	Requirement
Security	Passwords hashed, HTTPS har jagah, role-based access, har important change ka audit log
Privacy	Bachon ka data sirf school aur us ke parents ko nazar aaye; daily encrypted backup

Area	Requirement
Performance	Normal screens 2 second mein load hon; 500 users ek saath chalane par system na ruke
Availability	99.5% uptime; load shedding aur kamzor internet ke liye pages halka rakha jaye
Language	Poora interface English aur Urdu (RTL) mein; result card dono zubanon mein
Devices	Phone, tablet aur laptop par chale; Android par 2 GB RAM wale phone par bhi theek chale
Offline	Attendance aur marks entry bina internet ke save ho kar baad mein sync hon (P1)
Scalability	Ek hi system mein kai schools (multi-tenant), har school ka data alag
Data	Purana data Excel se import ho; kisi bhi waqt poora data export ho sake
Accessibility	Bara, saaf text aur kam training ke saath chalne wala simple UI

User flows

Teen flows school ke roz ke kaam ka zyada hissa cover karte hain.

1. Admission se pehli fee tak

1. Admin admission form bharta hai aur class/section chunta hai.
2. System student ID banata hai aur parent ka account (phone number se) khol deta hai.
3. Fee structure ke mutabiq pehla voucher khud ban jata hai.
4. Parent ko voucher aur login detail SMS/WhatsApp par milti hai.
5. Accountant payment record karta hai aur receipt PDF jati hai.

2. Roz ki attendance

1. Teacher phone par apni class kholta hai (sab default present).
2. Absent bachon par tap karke save karta hai.
3. System absent bachon ke parents ko alert bhejta hai.

4. Principal dashboard par school ki attendance % update ho jati hai.

3. Exam se result card tak

1. Admin exam schedule aur total marks set karta hai.
2. Har subject teacher apne marks enter karta hai.
3. Class teacher marks check karke result ko "final" karta hai.
4. System grades aur position nikalta hai aur result card PDF banata hai.
5. Principal approve karta hai, phir parents ko result publish hota hai.

Data model aur integrations

Main entities

Entity	Ahem fields	Rishta
School	naam, shehr, session, settings	Baqi sab is ke under (multi-tenant)
Student	ID, naam, DOB, class, section, status	Parent se linked; Class mein enrolled
Parent/Guardian	naam, phone, CNIC (optional)	Ek parent ke kai bache
Staff	naam, role, subjects, joining date	Teacher kai classes/subjects par
Class/Section	naam, class teacher, timetable	Students aur subjects ka group
Attendance	student/staff, date, status	Student ya Staff se linked
Fee Voucher	student, month, items, due date, status	Payments se linked
Payment	amount, method, date, receipt no.	Voucher se linked
Exam / Result	exam, subject, marks, grade	Student aur Subject se linked

Integrations

Service	Maqsad	Phase
SMS gateway (local provider)	Attendance, fee aur notice alerts	MVP

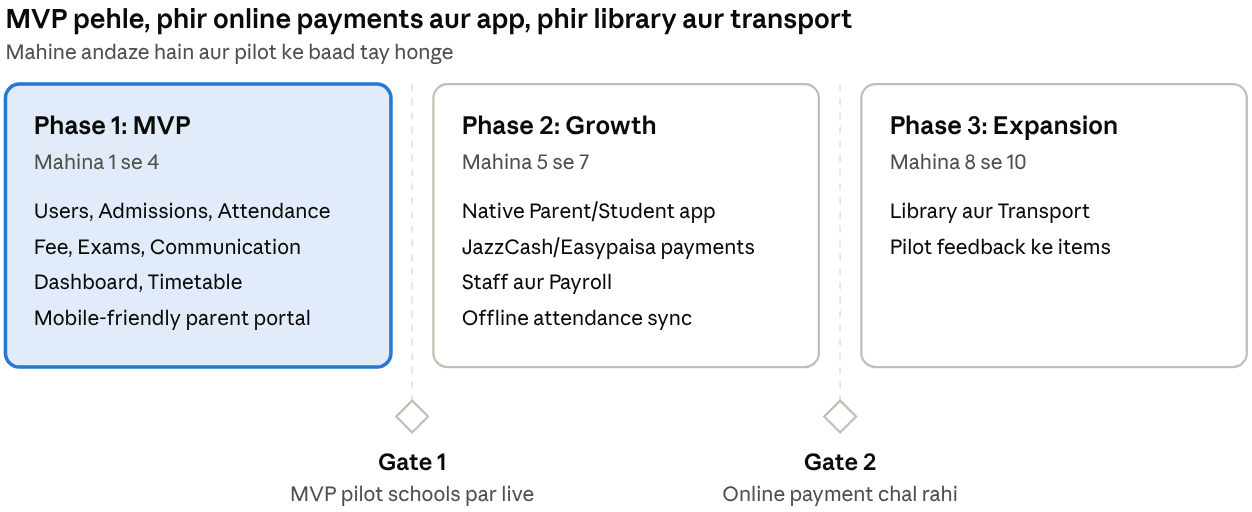
Service	Maqsad	Phase
WhatsApp Business API	Parents ko alerts aur receipts	MVP
JazzCash / Easypaisa	Online fee payment	Phase 2
Bank challan (1Bill ya bank file)	Fee payment reconciliation	Phase 2
Excel/CSV import-export	Purana data lana, reports nikalna	MVP

Success metrics

Ye targets launch ke 3 mahine baad pilot schools par napay jayenge; abhi ye shuruati andaze hain aur pilot ke baad tay honge.

Metric	Ab (andaza)	Target
Daily attendance mark karne ka waqt	15 minute (register)	5 minute se kam
Fee baqaya	Baseline pilot mein napa jayega	30% kam
Result card banane ka waqt	3 se 5 din	1 din se kam
Teachers jo har hafte system use karen	0%	85% se zyada
Parents jo mahine mein ek baar login karen	0%	60% se zyada
Support tickets per school per mahina	Naya product	10 se kam
Pilot schools jo launch ke baad bhi use jari rakhen	Naya product	80% se zyada

Roadmap aur release plan



roadmap · 3 phases, 2 gates

Har phase ke baad ek gate hai: agla phase tab shuru hoga jab pichla phase pilot schools mein sahi chal raha ho. Muddat (10 mahine) team size ke andaze par hai aur team tay hone ke baad dobara dekhi jayegi.

Risks, assumptions aur open questions

Risks

Risk	Asar	Bachao
Teachers naya system apnane se jhijkein	Attendance/marks data adhura rahe	Aasaan UI, 1 ghante ki training, pilot mein support
Internet ya bijli ki kami	Attendance aur marks entry mein ruka-wat	Halka UI, offline sync (P1)
Parent data ya bachon ke records leak hon	Bharosa aur qanooni masla	Role-based access, audit log, encrypted backup
SMS/WhatsApp ka kharcha zyada ho	School ki cost barh jaye	Alert settings school khud tay kare; kharcha report mein dikhe
Purana data ghalat ya adhura ho	Import mein ghaltiyan	Import ke waqt validation aur ghalat rows ki list

Assumptions

- Pehle target 100 se 1,500 students wale private schools hain.
- School ke paas kam az kam ek computer aur teachers ke paas smartphone hain.
- Parents ke paas WhatsApp ya SMS wala phone number hai.

Open questions

- [x] ~~Product ek school ke liye hai ya kai schools ko service ke taur par bechna hai (multi-tenant SaaS)?~~
- [x] ~~Pricing kaisi hogi: per student, per school ya yearly license?~~
- [x] ~~Kaunsa SMS/WhatsApp provider use hoga aur us ka kharcha kaun uthaye ga?~~
- [ ] Result card ka format kisi board (Federal, Punjab, Sindh, Cambridge) ke mutabiq chahiye?
- [ ] Pilot ke liye kaun se 2 se 3 schools tay hain?